

The insight theme on the ladder

Questions that evolve, on data that grows: how we ask, answer, check and hand
on what the data says

Analysis is cheap now; a reason to trust it is not

- An agent runs a hundred analyses in the time a person checks one.
- Each way an analysis fools itself has been measured, and none is rare.

failure	what was measured
Forking paths	29 teams, one dataset and question: odds ratios from 0.89 to 2.93 (Silberzahn 2018)
Subgroup claims	46 of 117 tested the difference; none of 5 attempts to confirm one did (Wallach 2017)
No replication	97% of 100 studies significant; 36% of their replications (Open Science 2015)
Self-checking	a language model rates its own output above others' (Panickssery 2024)

Each failure has a known remedy; the design puts every remedy on the path of every question.

Today we cannot say why an answer changed

- A register board holds one extract, its partitions and the questions, edited in place.
- A new extract, a fixed script and a new question tend to land in the same week.
- When an answer moves, nobody can name the one cause.

Today

- insights/<Dataset>-InsightBoard/: one extract
- questions and their code edited in place
- a new extract: a new board, or an overwrite



On the ladder

- a Prototype Block: the questions and their code, released p1, p2
- an insight Board: the data versions, a Job per pair
- a Job pins one release and one data version

Two things change, on two clocks: the questions and the data

- The questions change when we learn; the data changes when it arrives.
- A Job pins one of each, by path and hash, and is frozen when it closes.

Prototype

- the questions, their scripts, the cuts and the compare rule
- released $p1 \rightarrow p2 \rightarrow p3$; a person signs each
- lives in `work/bNN_<topic>_prototype/`

Dataset

- extracts by date, outside the Project
- versions $\langle d \rangle v1 \rightarrow \langle d \rangle v2$ build up
- a new one needs no release

Job

- one release $\times$ one data version
- `j03_p2_<d>v2` in `insights/bNN_<topic>/`
- frozen when closed; a change is a new Job

One change per Job, so every changed answer has one cause

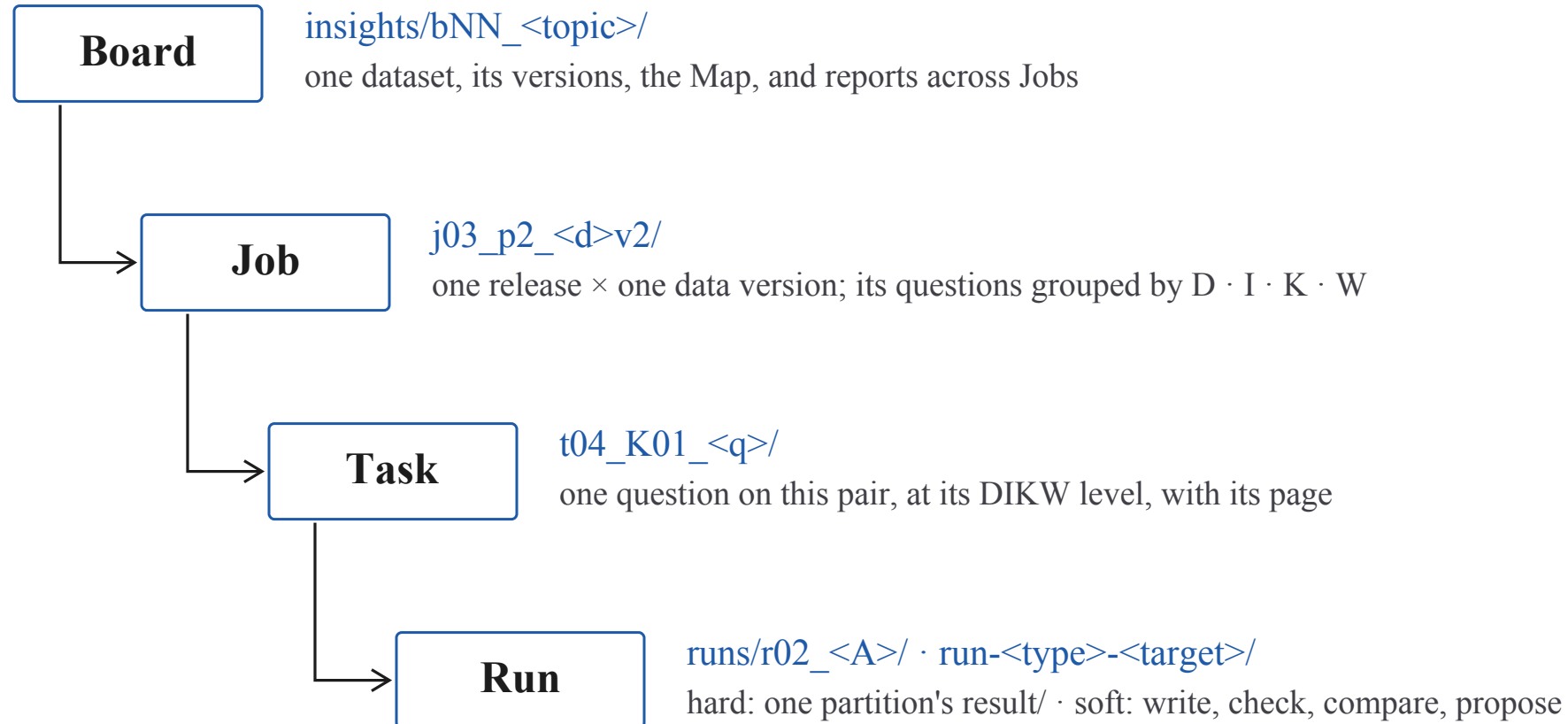
- Each step to the next Job moves the data or the code, never both.
- Backfill (new code on old data) is one more cell, not a special case.

	<d>v1	<d>v2	<d>v3
p1	j01 · start	j02 · data moved	—
p2	+ backfill p2 × <d>v1	j03 · code moved	j04 · data moved
p3	—	—	+ the next release

The Board's Map: data versions as columns, releases as rows. Jobs two clocks apart are never compared.

Four levels: Board, Job, Task, Run

- Every theme has these four levels; what each holds is insight's own.



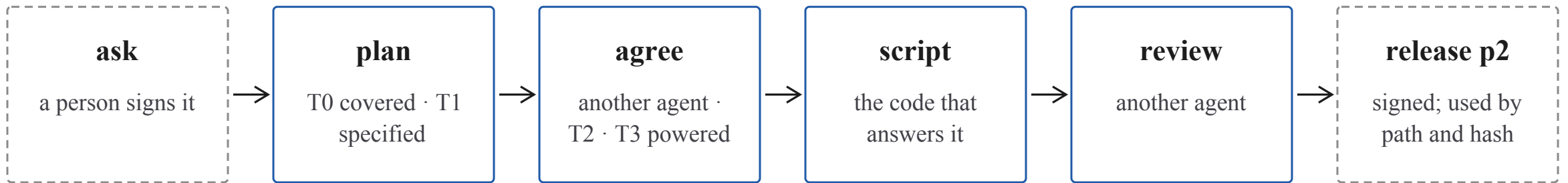
Each DIKW level may say more, and has one thing it may not say

- A question's level is set when it is asked, before the data, so a weak result cannot be promoted.

level	may say	may not say	must show
D · Data	counts, what was seen	any comparison	n, coverage, missingness
I · Information	rates, contrasts, segments	"because"	one test for each contrast
K · Knowledge	one claim, how sure, its rivals	advice	rivals ruled out; replicates (T8)
W · Wisdom	advice for Design	new evidence	the K claims it rests on; a signature

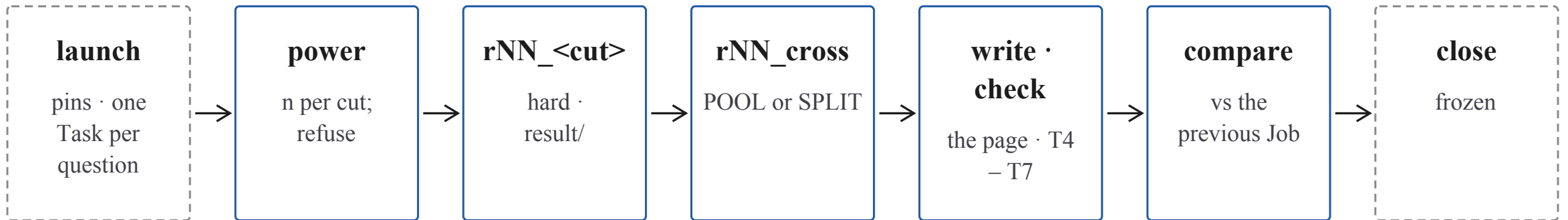
A question is planned, agreed and signed before any data is read

- The ask, the plan, the cuts, the power floors and the compare rule are fixed in a release.
- A different agent agrees the plan (T0 – T3), and a person signs the release.



Inside one Job: power first, then run, check, compare and propose

- Power is checked before any outcome is read, so a weak cut is refused and shown, never run.
- Each partition is its own hard Run; Cross runs one test of the difference.
- Compare and propose come after the checks, so a proposal rests on checked answers.



Four guardrails keep the answers honest

- None of them relies on the maker checking its own work.

Plan first

- the release is fixed before outcomes
- T0 – T3 gate the plan

Planned cuts

- filters and power floors live in the release
- a weak cut is refused and shown
- Cross tests the difference

Another agent checks

- T4 – T7 by an agent that did not write it
- T5: each sentence says what its result says

A person signs

- the release before the data
- the handoff after it
- nothing else leaves the Board

Every answer gets a status against the previous Job

- The rule for "held" is fixed in the release before any outcome, not by whoever reads the result.

status	means
new question	first asked in this release: not yet a finding
new finding	a question asked before now has an answer
held	the release's rule says it is the same
changed	it moved beyond the rule, and the moved clock is its cause
dropped · not comparable	answered before and not now · the question itself changed

Held: D counts within a tolerance; I the same direction; K the same sign and size; W its ground holds.

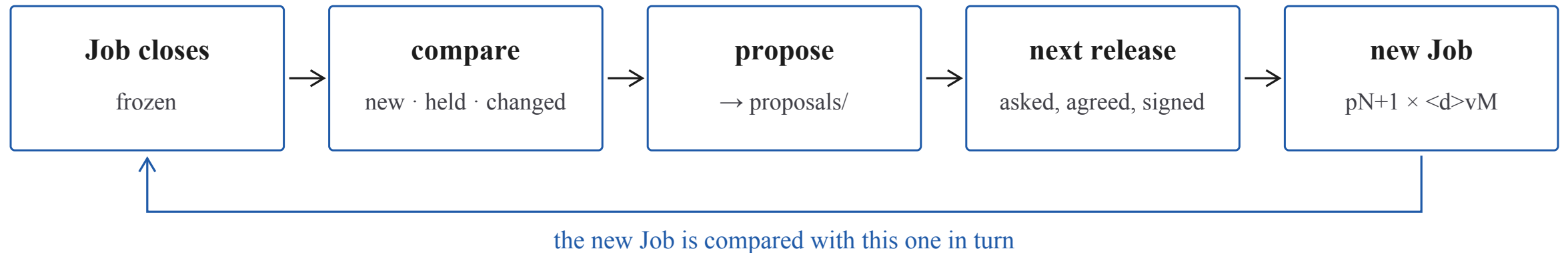
One question, four Jobs: what a naive pipeline would have shipped

- K01 asks whether <arm X> beats <arm Y> on <outcome>, split by <A> and .
- Illustrative only: placeholders, no data.

Job	moved	<A> vs 	a naive pipeline	this design
j01 p1 × v1	start	SPLIT: larger in <A>	ships "works better in <A>"	a K claim: W waits for T8
j02 p1 × v2	data	POOL	the effect "disappears"	changed; cause: data
j03 p2 × v2	code	 below power	old and new mixed	 refused, and shown
j04 p2 × v3	data	POOL	—	held: the split was noise

Comparing Jobs is how the questions grow

- Gaps and changed answers become proposals; proposals become the next release.
- A question added after seeing the data stays exploratory until the next data version confirms it.



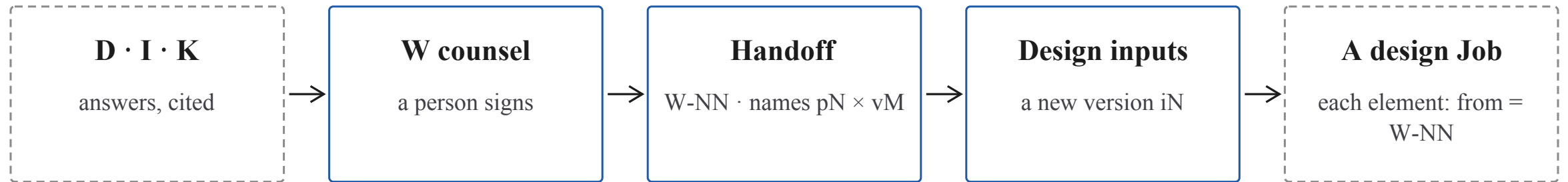
The Board reads across Jobs: a second D · I · K · W

- Readings use the Jobs' results, never raw data, and compare only Jobs one clock apart.

reading	asks	soft Run
Coverage	what was asked and answered, on which Job	run-coverage
Tracks	one question Job by Job; a trend over data versions	run-track-<q>
Consistency	each step held or changed, and its one cause	run-consistency-<jA>-<jB>
Findings	the Board's own questions, each citing a reading	run-report-<qNN>

Only signed Wisdom leaves, and it steers design

- D, I and K hand their answers up, not out.
- A signed Wisdom counsel names its Job, and enters the design Block as a new inputs version.



One workbench: the same six Spaces at every level

- Board, Job and Task each show the same six Spaces; what a Space holds follows the level.

Space	Board	Job	Task
Description	Map · Prototype · Dataset · Partitions	the release and the data version	Question · Plan · Data
Audience Report	Coverage · Tracks · Consistency · Findings	Question Work Report	need → Run → sentence
Work Details	the Jobs, their quick results	Job › Task › Run	Task › Run
Runs	add a version or a Job · readings	launch ... compare · propose	rNN · cross · write · check
Delivery	the signed handoff	its counsel, to the Board	W only

What it guarantees, and what it does not

- It narrows how an analysis can go wrong; it does not replace judgment about what to ask.

It guarantees

- every sentence traces to a Run and a release
- each change between Jobs has one cause
- tests and cuts were fixed before outcomes
- a person signed what left

It does not guarantee

- that the question was worth asking
- causal validity beyond what the data supports
- power in small data (it refuses instead)
- that the signer read carefully

It costs

- a Run per partition, plus cross, write, check
- a review and a signature per release
- coverage: refused cuts

The design is itself a bet, and we can measure it

- Planned, agreed, powered questions should replicate on the next extract more often than ad hoc ones.
- No study compares these methods head to head on one question; this would be the first.

measure	definition	read against
Replication (T8)	K answers held on the next data version	an agent's ad hoc analysis
Attributed changes	changed answers with one moved clock	100% by construction
Refusals	planned cuts refused for power	the coverage we give up
Time per Job	launch to close	the latency we pay

Where it stands

- Next: real Boards on the ladder, then the older register boards carry over.

Done

- the design: structure (s00), Board · Job · Task screens (s11 – s13), the Guide (s31)
- the workbench draws all three levels from a placeholder Board; 17 tests pass

In progress

- the run types and skills by level (s21)
- the first Board with a Prototype and data versions on disk

Open

- one Prototype per Board, or shared?
- when a release is cut
- does a signed handoff go stale?